

Security of Mission-Critical Machine-Type Communication

From 3GPP Standards to Emerging 6G Threats

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Abstract—Mission-critical machine-type communications (mc-MTC) support applications where delayed messages cause physical harm. Most security research evaluates mc-MTC via confidentiality and detection accuracy, implicitly assuming stable Ultra-Reliable Low-Latency Communication (URLLC) timing.

We systematise mc-MTC security across 91 papers and 14 3GPP specifications using a five-dimensional taxonomy, revealing that this assumption is fragile.

We identify a clear evaluation gap: 74% of papers assess standards-defined properties, 73% model only external adversaries, and none report deadline-miss rates. Analysis of TS 33.180 reveals limited formal verification of Mission Critical Services (MCX) protocol flows and no post-quantum cryptography consideration.

Empirically, using 5G New Radio (NR) Configured Grant in ns-3/5G-LENA, we show standards-compliant stressors inflate uplink latency by $\approx 3\times$ and cause $\approx 30\%$ deadline misses at 0.5 ms without protocol violation. The system fails *securely*.

We conclude mc-MTC security must treat availability, measured via latency distributions and deadline-miss rates, as a first-class security property.

Our code and experimental data are found at <https://github.com/UoS-SCCS/mc-MTC>.

I. INTRODUCTION

In August 2023, Meijer et al. revealed that TETRA’s TEA1 cipher, widely used by police and critical infrastructure, had been deliberately weakened to a 32-bit effective key length [1]. The flaw, introduced during standardisation and hidden for over 25 years by proprietary specifications, highlights the risks in today’s migration from closed narrowband systems to 3GPP-based 5G mission-critical services.

Mission-critical machine-type communications (mc-MTC) support applications where communication failure can cause physical or economic harm, including industrial control, smart grids, cooperative driving, and public safety. Unlike conventional systems, mc-MTC prioritises availability and integrity over confidentiality, since delayed or corrupted control traffic can destabilise systems before data leakage causes harm [2].

Existing research explores physical-layer authentication [3], schedule randomisation [4], zero-touch slice protection [5], lightweight cryptography, and ML-based detection [6]. Yet these approaches are usually evaluated in isolation, assume stable timing conditions, and focus mainly on external attackers. Meanwhile, 3GPP Mission Critical Services (MCX)—MCPTT, MCVideo, and MCData—secured under TS 33.180 [7] are increasingly deployed, but no unified systematisation of mc-MTC security currently exists.

Thesis. This work argues that mc-MTC security research treats latency as a system constraint rather than a security metric. We show this assumption is fragile. Using a reproducible 5G NR Configured Grant case study, we demonstrate that standards-compliant stressors, including insider workload surges and operator policy changes, can increase mean latency by $\approx 3\times$ and cause $\approx 30\%$ deadline misses at 0.5 ms without cryptographic compromise or protocol violation. The system can therefore *fail securely*: availability remains vulnerable when scheduler behaviour and control-plane policy are ignored.

Scope. We systematise 3GPP mission-critical communications, covering MCX and TS 33.180, URLLC bearer-layer security, and sidelink/off-network extensions in TS 33.503 [8]. MCX is the primary focus due to its maturity and deployment urgency, while industrial URLLC, military tactical, and V2X systems are treated as adjacent domains with shared threat characteristics.

Contributions. This work makes five contributions:

1. Availability-centric reframing. We challenge five implicit assumptions in the literature, arguing that mc-MTC security must treat availability as a first-class property, with latency distributions and deadline-miss rates as primary security outcomes.

2. Five-dimensional taxonomy. We classify mc-MTC security along five dimensions (stack layer, objective, adversary, resource lever, key performance indicator (KPI)). Applying

this to 91 papers and 14 3GPP specifications assembled via structured search following PRISMA reporting guidelines, we derive a quantitative coverage map (Figure 1). We find that privacy is largely unstudied, and no paper reports deadline-miss rates (Table II).

3. TS 33.180 primary analysis. We directly analyse 3GPP’s TS 33.180 version 20.0.0. We identify normative key escrow (Section 10.2.4), a 14+ key hierarchy, no references to post-quantum cryptography, and the absence of formal verification for 11 of 13 MCX protocol flows (Table I).

4. Empirical demonstration. Using 5G NR Configured Grant in ns-3/5G-LENA, we demonstrate that standards-compliant stressors (T_1, T_2) inflate mean latency by $\approx 3\times$ and induce $\approx 30\%$ deadline misses at 0.5 ms. (All of our code and artifacts available at <https://github.com/UoS-SCCS/mc-MTC>).

5. Evidence-driven research agenda. We synthesise eleven urgent gaps covering MCX formal verification, post-quantum migration for MIKEY-SAKKE (dependent on an Internet Engineering Task Force (IETF) Request for Comments (RFC) dormant since 2012), scheduler-aware threats, and policy-safe configuration envelopes.

II. WHAT THE COMMUNITY ASSUMES

The mc-MTC security literature is rich in mechanisms and threat models, yet it rests on several implicit assumptions that constrain how the field reasons about mission-critical risk. We articulate five such assumptions, validate them against our corpus (Section IV-A), and expose their divergence from system-level behaviour. We validate these observations empirically in Section V.

A. A1: Standards Compliance Implies System Security

Literature frequently equates adherence to 3GPP security specifications (e.g., mutual authentication, ciphering, and slice isolation) with mission-critical robustness. Prior works [9]–[11] evaluate 5G security primarily via standards-defined mechanisms. This conflates *mechanism provision* with *mission assurance*. A system can perfectly implement specified mechanisms yet fail its mission if availability breaches control-application latency thresholds.

In our 91-paper corpus, 67 (74%) evaluate security against standards-defined properties (confidentiality, authentication, integrity). Only 10 (11%) measure impact on latency distributions or deadline-miss rates, the true KPIs of mc-MTC availability. Section V demonstrates how standards-compliant configurations can induce availability collapse without mechanism failure.

B. A2: Threats Originate from External or Rogue Actors

The dominant wireless threat model assumes malicious external entities or rogue infrastructure [12]–[15]. In our corpus, 66 papers (73%) model exclusively external adversaries; only 12 (13%) consider non-external threat models (insiders, privileged operators, or cross-slice actors), and none model authenticated, protocol-compliant endpoints as availability stressors. Yet, in mission-critical deployments, already-admitted

devices (e.g., PLCs, sensors, body-worn cameras) may be misconfigured or application-compromised while remaining network-authenticated. The availability degradation emerges from structural resource contention, not identity abuse.

C. A3: Scheduler Efficiency Provides Robustness

OFDMA schedulers are widely considered more efficient than TDMA alternatives, implicitly assuming greater resilience under stress [4]. While OFDMA achieves the lowest mean latencies under nominal load (Section V), nominal efficiency does not guarantee robustness under structural resource reduction. No corpus paper evaluates scheduler behaviour under adversarial or policy-induced resource contraction; scheduler selection is uniformly treated as a performance optimisation, not a security decision.

D. A4: Policy Knobs Are Operational, Not Security-Relevant

Operator-level parameters (e.g., DL/UL symbol allocation, slice Quality of Service (QoS) profiles, scheduling policy selection) are treated in the standards and in practice as performance-tuning tools [16]. The O-RAN architecture delegates these decisions to the near-RT RIC and SMO [17], which are assumed to operate within a trusted domain. In our corpus, no paper models configuration parameters as part of the attack surface. The closest related work discusses O-RAN security conceptually [18] but provides no empirical assessment of availability impact from policy changes.

For mission-critical deployments, this assumption is hazardous. A DL/UL split change from 1D13U to 9D5U, for instance, is operationally legitimate and standards-compliant, yet it fundamentally alters the latency distribution of Configured Grant transmissions. Policy guardrails, safe operating envelopes, and automated change-control validation must be treated as integral to the security model rather than excluded from it.

E. A5: Security Overhead Is the Primary Latency Risk

A common narrative frames cryptographic processing cost as the principal threat to URLLC latency budgets. This drives research into finite-blocklength secrecy, low-overhead PLA, and lightweight cryptography [2], [3], [19]. Post-quantum cryptography (PQC) raises further concern: lattice-based Identity-Based Encryption (IBE) candidates produce ciphertexts $100\times$ larger than MIKEY-SAKKE, with uncertain implications for Session Initiation Protocol/Session Description Protocol (SIP/SDP) message sizes [20].

While valid, these concerns focus on *additive* overhead from security mechanisms while overlooking *structural* resource sensitivity. Severe latency degradation can occur without introducing any additional security mechanism. As demonstrated in Section V, our stressors add no cryptographic overhead yet produce $\approx 3\times$ mean latency inflation. The dominant fragility arises from resource allocation, queueing dynamics, and transport-block sizing (factors that exist independently of any security mechanism).

F. Synthesis

These five assumptions share a common thread: they locate mc-MTC security risk in *adversarial action* (external attackers, cryptographic breaks, protocol violations) and evaluate defences through *mechanism-centric metrics* (detection accuracy, secrecy rate, cryptographic strength). This framing is incomplete. It ignores endogenous stress, policy-induced resource contraction, and the structural coupling between scheduling determinism and availability. As our corpus demonstrates (74% evaluate standards-defined properties, 73% model external adversaries, and only 11% measure latency distributions), mc-MTC security must be reframed around structural availability and configuration safety rather than solely adversarial attack resistance.

III. BACKGROUND

A. Legacy Mission-Critical Systems

Legacy mc-MTC systems illustrate the risks of closed security design. TETRA's proprietary TEA ciphers were reverse engineered by Meijer et al. [1], revealing effective 32-bit security and a decryption oracle (CVE-2022-24401). P25 suffered repeated implementation failures, including plaintext transmission caused by optional encryption and manual key handling [21]. TETRAPOL remains proprietary with no public analysis. Across all three systems, limited scrutiny enabled decades-long vulnerabilities, motivating our examination of whether similar structural weaknesses now exist in 3GPP MCX.

B. 3GPP Mission Critical Services

3GPP MCX replaces legacy narrowband systems with broadband services. TS 23.280 [22] defines MCPTT, MCVideo, and MCDATA alongside identity, group-management, and MIKEY-SAKKE [23] key-management servers. TS 33.180 [7] specifies layered authentication, a hierarchy of 14+ key types, and end-to-end media protection across on-network, off-network, and interconnection deployments.

Two design features are notable. First, TS 33.180 Section 10.2.4 explicitly enables lawful decryption through KMS key escrow. Second, an unresolved Editor's Note in Section 9.3.1 questions how MCPTT encryption should operate during VPLMN roaming; the note persists from Rel-15 to Rel-20, indicating unresolved ambiguity in a deployed standard.

C. URLLC and Configured Grant Scheduling

mc-MTC depends on URLLC, targeting packet error rates of 10^{-5} – 10^{-9} and sub-1 ms latency [24]. TS 22.104 defines survival times as low as 0 ms, so missed packets become safety failures. 5G uplink scheduling includes dynamic scheduling, semi-persistent scheduling (SPS), and Configured Grant (CG) [16]. CG pre-allocates uplink resources via RRC signalling, reducing latency and jitter by removing per-packet control overhead. However, its determinism tightly couples traffic load and latency:

App. Load → TB Allocation → Queueing → Latency Distribution (1)

Fixed transport-block (TB) sizes mean excess payload spills into later CG occasions, causing RLC queue growth. Similarly, reducing uplink symbols decreases transmission opportunities and inflates tail latency. Both effects degrade availability without protocol violation; Section V quantifies them.

IV. OUR FIVE-DIMENSION SECURITY FOR MC-MTC

Existing mc-MTC security research is fragmented across communities that rarely intersect: protocol analyses focus on Dolev–Yao authentication models [10], PHY-layer work optimises secrecy under finite blocklengths [19], and standards bodies publish specifications without independent adversarial evaluation [7]. So, we propose a five-dimensional taxonomy capturing *where* attacks occur, *what* they target, *who* performs them, *which lever* they exploit, and *how* they affect KPIs.

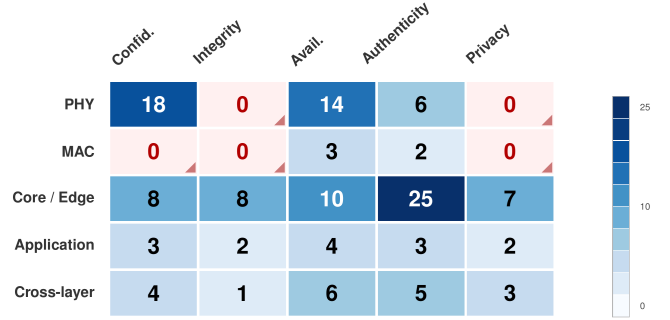


Fig. 1: Coverage map of the 91-paper corpus across stack layer and security objective. 52% of cells contain ≤ 3 papers; red cells denote unstudied intersections.

Unlike CIA-centric frameworks, our taxonomy prioritises availability, reflecting the inverted priorities of mc-MTC [2]. Figure 1 summarises the resulting 91-paper and 14-specification corpus.

A. Corpus Assembly and Classification

We searched IEEE Xplore, ACM DL, SpringerLink, and arXiv from Rel-13 (2016) to January 2025 using PRISMA guidelines [25]. Queries combined mission-critical terms (e.g., MCX, MCPTT, mc-MTC), 5G/URLLC terms, and security terms. Forward/backward citation analysis added further work from venues such as Wiley and MDPI.

After deduplication and screening, papers were retained if they: (i) addressed mission-critical security properties, (ii) targeted the 3GPP ecosystem, and (iii) contained technical contributions or explicit threat analysis. We supplemented these with 14 3GPP specifications [7], [8], [16], [22], [24], [26]–[34] and seminal works [1]–[3], [10].

Final corpus: 91 papers and 14 specifications. We classified these 91 papers¹ independently across all five dimensions

¹See Appendix A for the full list.

mentioned below. We excluded of most pre-Rel-13 LTE-MTC work, as well as some literature linked to MTC but irrelevant here [35]–[37].

B. Dimension 1: Stack Layer

PHY layer. There is research focused on jamming, spoofing, and PLA [3], [40], plus finite-blocklength secrecy coding [19]. Most assumes stable scheduling and benign channels.

MAC layer. Only 5 papers study deterministic scheduling and timing attacks [4]. None model endogenous load amplification like our stressor T_1 .

Control/core layer. 35 papers address slicing and DDoS threats [5], while AdaptiveMobile and NSA/CISA identify unresolved cross-slice risks [37], [41].

Application layer. Only 8 papers study replay, false-command injection, or MCX signalling. MCPTT floor control and MIKEY-SAKKE remain formally unverified (Table I).

C. Dimension 2: Security Objective

mc-MTC prioritises availability and integrity over confidentiality [2]. Across the corpus, confidentiality appears in 36% of papers, authenticity in 45%, availability in 41%, integrity in 12%, and privacy in 13%.

However, only 10 papers evaluate latency distributions or deadline-miss behaviour, despite TS 22.280 availability requirements [29]. MCX privacy is also underexplored: TS 33.180 specifies only limited SIP-level protections [7], and only 4 papers examine MCX-specific privacy [42]. Our stressors T_1/T_2 demonstrate severe availability collapse without confidentiality failure.

D. Dimension 3: Adversary Model

External adversaries dominate the literature (73%), including jammers, spoofers, and DDoS attackers [3], [5], [12], [13].

Authenticated insiders are rarely studied despite being able to degrade availability through legitimate traffic amplification [43]; only 3 papers consider them.

Operator-level actors can alter scheduler or slicing policy without violating standards [16], [17], yet almost no empirical work evaluates resulting availability impact [18], [44].

Cross-slice adversaries exploit virtualisation boundaries [33], [37], [41]; only 6 papers consider them, none in mc-MTC contexts.

E. Dimension 4: Resource Lever

Radio and scheduling. Most work studies jamming and interference [3], [40], but not policy-driven effects such as DL/UL reallocation. Deterministic CG/SPS scheduling creates exploitable predictability [4].

Traffic and computation. Protocol-compliant traffic growth causes RLC queue inflation. In stressor T_1 , increasing PDCP SDU size from 10 B to 300 B raised latency from ≈ 0.32 ms to ≈ 1.5 ms (Table III). PQC overhead further threatens timing budgets [20].

Virtualisation boundaries. MEC and slicing introduce isolation challenges. Nencioni et al. [45] show migration delays incompatible with mc-MTC, while AdaptiveMobile [41] and TR 33.880 highlight unresolved cross-slice risks.

F. Dimension 5: KPI Impact

Latency and deadline misses. TS 22.104 defines sub-millisecond URLLC targets [24], yet only 11% of papers evaluate latency distributions and none report deadline-miss rates. Our stressor T_2 induces $\approx 30\%$ misses under standards-compliant policy changes.

Reliability and ML accuracy. BLER/PER dominate PHY evaluations [3], [19], while ML-based work reports detection accuracy [5], [6] but rarely quantifies inference-latency costs.

Cryptographic strength. Formal proofs dominate 36% of the corpus [10], [39], but abstract away scheduler and configuration dynamics that determine availability.

Synthesis. Table II reveals a major evaluation mismatch: formal proofs and accuracy metrics dominate, while latency distributions and deadline-miss rates—the KPIs most relevant to mc-MTC safety—are almost entirely absent.

V. CASE STUDY: CONFIGURED GRANT UNDER STRESS

To ground our availability-centric systematisation in measurable evidence, we examine 5G NR *Configured Grant* (CG) uplink scheduling under two realistic stressors. *These experiments are demonstrative evidence for the thesis of this work.*

Note: All our code and artifacts for this can be found at <https://github.com/UoS-SCCS/mc-MTC>.

Configured Grant is widely adopted for periodic mc-MTC traffic because it removes dynamic grant signalling, reducing control-plane latency and jitter. Unlike dynamic scheduling, CG pre-allocates uplink resources, providing deterministic transmission opportunities. This determinism is the source of its efficiency.

Further, it is also the source of its fragility. By eliminating per-packet negotiation, CG creates a tightly coupled system (Equation (1), Section III-C). Our goal is not to exhaustively benchmark schedulers, but to expose how standards-compliant stressors propagate through this structural coupling into mission-critical KPIs.

A. Experimental Setup

We build on the public 5G-LENA/ns-3 implementation of NR Configured Grant. Unless otherwise stated, the environment is: single cell; carrier 3.75 GHz; bandwidth 20 MHz; numerology $\mu = 1$; 15 UEs with periodic uplink traffic; HARQ disabled; fixed UL MCS (`StartingMcsUl=12`); simulation time 10 s; one representative random seed. We evaluate four schedulers available in 5G-LENA: 5GL-TDMA, 5GL-OFDMA, Sym-OFDMA, and RB-OFDMA. Baseline slot format is 1D13U.

The primary metric is one-way PDCP→RLC latency. We report mean, min, and max latency for both stressors. For T_2 we additionally report the latency CDF and deadline-miss rate at 0.5 ms. This configuration isolates scheduler and resource effects without cross-cell interference, enabling structural causality to be observed.

Scope and limitations. The study is single-cell, single-numerology, CG-only with HARQ disabled. These choices

TABLE I: Formal verification status of MCX-specific protocol flows. 5G-AKA and EAP-AKA' are included for comparison; all MCX-specific flows remain unverified.

Protocol / Flow	Specification	Verified?	Reference / Note
5G-AKA	TS 33.501	Yes	Basin et al. [10]
EAP-AKA'	TS 33.501	Yes	Cremers and Dehnel-Wild [38]
† MCPTT private/group setup	TS 24.379 / TS 33.180	No	—
† MCPTT floor control	TS 24.380 / TS 33.180	No	—
† MCVideo / MCDATA control	TS 24.581 / TS 24.582	No	—
† MIKEY-SAKKE transport	RFC 6509 / TS 33.180	No	Primitive proven [39]; composition unverified
† GMK / PCK / CSK distribution	TS 33.180	No	—
† Off-network / IOPS security	TS 33.180	No	Editor's Note persists in §9.3.1
† MC Gateway interworking	TS 33.180 Annex L	No	—

TABLE II: Primary evaluations across the corpus. Deadline-miss rate absent despite importance to mc-MTC safety.

Metric	Papers	%
Formal proof / crypto strength	33	36%
Detection accuracy	20	22%
Secrecy rate / capacity	16	18%
Latency distributions	10	11%
BLER / PER	8	9%
Deadline-miss rate	0	0%
<i>Total papers</i>	91	

isolate the scheduler/policy effects under examination. Multi-cell dynamics, HARQ retransmissions, and mixed numerology would introduce additional latency variance; we leave these extensions to future work and provide all simulation scripts for replication.

B. Stressor T_1 : Insider Workload Surge

T_1 models an authenticated insider increasing application-layer payload size while remaining fully protocol-compliant. UE population, radio parameters, and CG configuration remain fixed. We sweep PDCP SDU sizes $\{10, 150, 300\}$ B under the baseline 1D13U slot format.

Under our taxonomy (Section IV), T_1 is an application-to-MAC phenomenon (Dimension 1): it originates at the application layer but manifests through MAC scheduling and buffering. Its security objective is availability (Dimension 2); its adversary model is an authenticated insider (Dimension 3); its resource lever is traffic load amplification (Dimension 4); and it is measured via latency inflation (Dimension 5).

Table III shows uplink latency increasing monotonically with payload size. At 10 B, Sym-/RB-OFDMA achieve the lowest means (≈ 0.324 ms) and maxima (≤ 0.457 ms), outperforming 5GL-TDMA (0.486 ms mean) and 5GL-OFDMA (0.564 ms mean). As payloads grow, this advantage narrows because TB capacity is exceeded more frequently, causing RLC segmentation and queueing. At 300 B, mean latency rises to ≈ 1.50 – 1.55 ms, with maxima reaching 2.56–2.85 ms.

The degradation arises from structural amplification, not adversarial radio behaviour. Excess payload spills into later Configured Grant (CG) occasions, increasing queue depth and

TABLE III: T_1 (insider workload surge): UL PDCP→RLC latency vs. payload size at 20 MHz, $\mu = 1$, 15 UEs, 1D13U. Sym-OFDMA at 300 B is omitted due to a known assertion in the current 5G-LENA OFDMA scheduler branch.

Scenario	Mean (ms)	Min (ms)	Max (ms)
TDMA ($p=10$ B)	0.486	0.207	0.814
TDMA ($p=150$ B)	0.843	0.243	1.457
TDMA ($p=300$ B)	1.552	0.314	2.850
5GL-OFDMA ($p=10$ B)	0.564	0.564	0.564
5GL-OFDMA ($p=150$ B)	0.931	0.564	1.564
5GL-OFDMA ($p=300$ B)	1.498	0.564	2.564
Sym-OFDMA ($p=10$ B)	0.324	0.207	0.457
Sym-OFDMA ($p=150$ B)	0.886	0.243	1.529
RB-OFDMA ($p=10$ B)	0.324	0.207	0.457
RB-OFDMA ($p=150$ B)	1.017	0.421	1.421
RB-OFDMA ($p=300$ B)	1.517	0.421	2.564

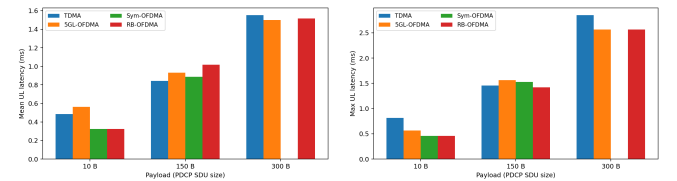


Fig. 2: T_1 (insider workload surge) under NR Configured Grant with $\mu = 1$, BW=20 MHz, 15 UEs, slot format 1D13U, UL MCS 12. Increasing PDCP SDU size (10B→150B→300B) raises latency across all schedulers.

inflating tail latency. Since no rogue device, cryptographic failure, or protocol violation is involved, the system *fails securely*. Deterministic scheduling therefore converts endogenous load growth into an availability risk.

Link to assumptions A2 and A5. T_1 shows that availability degradation can result from authenticated insiders and protocol-compliant load amplification, challenging the assumptions that threats are mainly external (A2, Section II-B) and that cryptographic overhead dominates latency risk (A5, Section II-E).

TABLE IV: T_2 (privileged policy change): configured-grant UL latency under baseline (1D13U) vs. 9D5U at 20 MHz, $\mu = 1$, 15 UEs, 10 B payload.

Scheduler	Scenario	Mean (ms)	Min (ms)	Ma
TDMA	baseline (1D13U)	0.486	0.207	
TDMA	attack (9D5U)	1.529	0.493	
5GL-OFDMA	baseline (1D13U)	0.564	0.564	
5GL-OFDMA	attack (9D5U)	0.998	0.564	
Sym-OFDMA	baseline (1D13U)	0.324	0.207	
Sym-OFDMA	attack (9D5U)	1.021	0.493	
RB-OFDMA	baseline (1D13U)	0.324	0.207	
RB-OFDMA	attack (9D5U)	0.924	0.493	

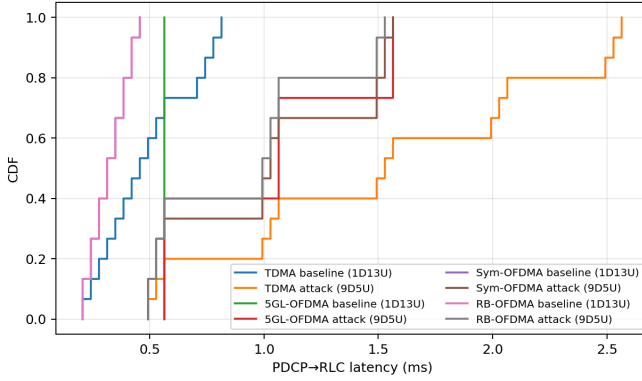


Fig. 3: Latency CDF (Configured Grant, $\mu = 1$, 20 MHz, 15 UEs). Baselines (1D13U) show low latencies for Sym-/RB-OFDMA, while a policy change to 9D5U shifts the tail rightward (higher P95/P99 and max).

C. Stressor T_2 : Policy-Induced UL Reduction

T_2 models a standards-compliant operator policy change. We modify the DL/UL split from 1D13U to 9D5U, reducing the uplink symbol budget while keeping all other parameters unchanged. This induces a second structural coupling, distinct from Equation (1):

$$\text{UL Budget Reduction} \rightarrow \text{Tx Deferral} \rightarrow \text{Queue Growth} \rightarrow \text{Deadline Misses} \quad (2)$$

Under our taxonomy, T_2 is a control-policy-to-MAC phenomenon (Dimension 1). Its security objective is availability (Dimension 2); its adversary model is a privileged operator-level actor (Dimension 3); its resource lever is UL symbol budget reduction (Dimension 4); and it is measured via latency distribution shift and deadline-miss rate (Dimension 5).

Table IV, Figure 3, and Figure 4 quantify the effect of shrinking the UL symbol budget. Mean latency multiplies by $\sim 3.1\times$ for TDMA (0.486 $\rightarrow$ 1.529 ms), $\sim 1.8\times$ for 5GL-OFDMA (0.564 $\rightarrow$ 0.998 ms), $\sim 3.1\times$ for Sym-OFDMA (0.324 $\rightarrow$ 1.021 ms), and $\sim 2.9\times$ for RB-OFDMA (0.324 $\rightarrow$ 0.924 ms). The CDFs shift right with heavier tails,

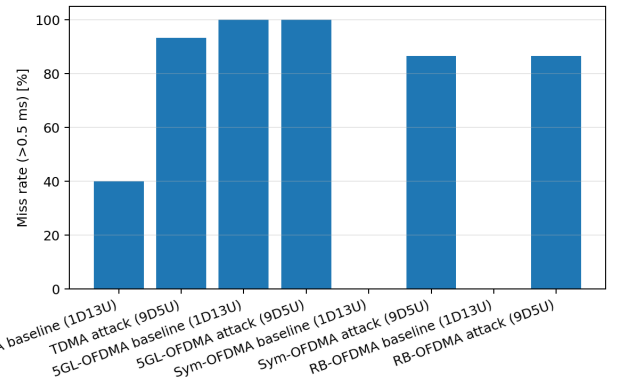


Fig. 4: Deadline-miss rate at 0.5 ms (Configured Grant). The 9D5U policy induces $\approx 30\%$ misses for Sym-/RB-OFDMA, whereas baselines show $\approx 0\%$.

and deadline misses at 0.5 ms jump to $\approx 30\%$ for both OFDMA schedulers while baselines remain near 0%.

Availability collapse emerges from permissible policy change.

Link to assumptions A1, A3, and A4. T_2 contradicts the assumptions that standards compliance guarantees safety (A1, Section II-A), that scheduler efficiency ensures robustness (A3, Section II-C), and that DL/UL configuration knobs are merely operational (A4, Section II-D).

D. Discussion

Table V links taxonomy dimensions, violated assumptions, and empirical evidence. Both stressors preserve trust boundaries, cryptography, and protocol compliance, yet still cause severe degradation in mission-critical KPIs. The shared pathway is:

Security/Policy Decision $\rightarrow$ Resource Lever $\rightarrow$ Scheduler Dynamics $\rightarrow$ Latency Distribution Shift $\rightarrow$ Deadline-Miss Inflation

In T_1 , the lever is endogenous load amplification; in T_2 , uplink-symbol reduction. In both cases, deterministic scheduling transforms resource imbalance into queue growth and tail-latency inflation. Thus:

Deterministic scheduling reduces variance under ideal load but amplifies fragility under constrained resources.

This fragility extends beyond Configured Grant to SPS, grant-free access, and other deterministic URLLC MAC schemes.

Five implications follow:

- 1) *Threats map directly to KPIs*: both stressors primarily degrade availability.
- 2) *Operator policy is a security surface*: T_2 shows standards-compliant configuration changes can induce CG starvation (Gap G7, Section VII-G).
- 3) *Endpoint discipline matters*: T_1 motivates rate limiting, payload budgeting, and adaptive TB sizing.
- 4) *Scheduler efficiency is not immunity*: OFDMA delays collapse but still suffers severe tail inflation under reduced UL budgets.

TABLE V: Our taxonomy’s dimensions and violated assumptions instantiated on T_1/T_2 .

Scenario	D1: Layer	D2: Goal	D3: Adversary	D4: Resource Lever	D5: KPI Impact	Assumptions Violated
T_1	App.→MAC	Availability	Auth. insider	Load amplification	Mean/max inflation (Table III)	A2, A5
T_2	Policy→MAC	Availability	Priv. operator	UL symbol reduction	CDF shift + miss@0.5 ms (Table IV, Figs. 3–4)	A1, A3, A4

5) *Effects are operationally actionable*: $\approx 3\times$ mean inflation and $\approx 30\%$ deadline misses provide concrete thresholds for monitoring and SLA enforcement.

The system can therefore *fail securely*: no primitive breaks, no unauthorised access occurs, yet safety-critical timing collapses.

Most prior work optimises detection accuracy, secrecy rate, or cryptographic strength [3], [19], [40]. Our results show the dominant failure mode in mc-MTC is instead latency-distribution collapse:

Under what endogenous stresses does latency collapse despite full standards compliance?

Synthesis. Security and policy decisions shape scheduler dynamics; scheduler dynamics shape latency distributions; latency distributions determine availability. mc-MTC security must therefore be co-designed with scheduling and resource allocation under explicit deadline-miss constraints.

VI. RELATED WORK

Prior work covers only fragments of the space addressed in this paper. Existing surveys fall into three categories: mc-MTC/URLLC surveys, 5G security systematisations, and mission-critical standards analyses.

mc-MTC and URLLC surveys. Mohammed et al. [2] provide the canonical overview of mc-MTC requirements and technologies, but treat security only as an open challenge without adversary models or empirical analysis. Mahmood et al. [46] discuss 6G MTC enablers and briefly mention security, but omit MCX protocols and key management. Nencioni et al. [45] survey MEC for URLLC and identify security as immature, yet limit their scope to edge computing. None model availability as a latency-distribution property or evaluate deadline-miss behaviour.

5G security systematisations. Eleftherakis et al. [47] present the closest related work, analysing 5G protocol and privacy vulnerabilities. However, MCX, TS 33.180, and MIKEY-SAKKE are absent, and availability is not treated as a security property. Ahmad et al. [9] survey standards-defined 5G security mechanisms but do not examine mc-MTC mission assurance. Basin et al. [10] and Cremers and Dehnel-Wild [38] formally verify 5G-AKA and EAP-AKA', while Hussain et al. [48], RANsacked [12], and 5Ghoul [14] demonstrate practical cellular vulnerabilities without considering mission-critical availability requirements.

Mission-critical standards analyses. Meijer et al. [1] demonstrate the risks of unverified mission-critical standards through the TETRA disclosure. NSA/CISA [37] and ENISA [11] assess 5G slicing and threat landscapes but do not analyse TS 33.180 or MCX protocols directly. Verchyk and

Sep'ulveda [20] evaluate post-quantum IBE candidates relevant to MIKEY-SAKKE replacement, but not the broader dependency and migration chain.

Positioning of this work. This paper is, to our knowledge, the first to combine: (i) primary-source analysis of TS 33.180 and its verification status; (ii) a quantitative five-dimensional taxonomy; (iii) empirical demonstration of availability collapse under standards-compliant stressors; and (iv) treatment of availability as a first-class security property measured through latency distributions and deadline-miss rates. Table VI summarises the comparison.

VII. RESEARCH GAPS

Our systematisation identifies eleven urgent gaps spanning protocol verification, availability modelling, and emerging 6G threats.

A. G1: No Formal Verification of MCX

Of 13 MCX protocol flows, 11 lack symbolic or computational analysis (Table I), including MIKEY-SAKKE transport, group keys, and floor control. While Sakai–Kasahara IBE has an IND-CCA2 proof [39], its deployment within TS 33.180 [7] alongside SIP, OAuth, and group key hierarchies has never been analysed, unlike 5G-AKA [10], [38]. *Target*: formal methods. *Needed*: symbolic MCPTT and MIKEY-SAKKE models.

B. G2: No Post-Quantum Migration Path

MIKEY-SAKKE [23] relies on pairing-based ECC vulnerable to Shor, yet TS 33.180 contains no PQC references. Migration depends on a dormant RFC and unscheduled SA3 work. Existing PQC-IBE proposals fail current MCX latency and message-size constraints [20]. *Target*: applied crypto, IETF, SA3. *Needed*: low-latency PQC-IBE for SIP/SDP.

C. G3: Declining MCX Interoperability

ETSI Plugtests report worsening interoperability as Rel-17 features are added [35], [36], especially around KMS and MIKEY-SAKKE interpretations. *Target*: ETSI, certification bodies. *Needed*: adversarial interoperability testing.

D. G4: MCX Privacy and Metadata Exposure

TS 33.180 defines only minimal privacy protections [7], while KMS escrow enables retrospective decryption. Only four corpus papers address privacy [42], none analysing MCX metadata flows. *Target*: privacy engineering. *Needed*: metadata leakage analysis.

TABLE VI: Comparison with related work; • = covered in depth; ○ = mentioned but not analysed; — = absent.

Work	MCX / TS 33.180	MIKEY-SAKKE	Formal Verif.	Avail. as KPI	Empirical	Taxonomy	PQC
Mohammed et al. [2]	○	—	—	—	—	—	—
Mahmood et al. [46]	—	—	—	○	—	—	○
Nencioni et al. [45]	—	—	—	○	—	—	—
Ahmad et al. [9]	—	—	—	—	—	○	—
Eleftherakis et al. [47]	—	—	•	—	•	•	—
Basin et al. [10]	—	—	•	—	—	—	—
Meijer et al. [1]	—	—	—	—	•	—	—
NSA/CISA [37]	○	—	—	○	—	—	—
ENISA [11]	○	—	—	○	—	—	—
This work	•	•	○	•	•	•	○

E. G5: Off-Network and IOPS Security

Off-network MCX and IOPS rely on weaker trust assumptions and unresolved roaming encryption issues in TS 33.180. Recent sidelink vulnerabilities further expand the attack surface [15]. *Target:* protocol security, D2D. *Needed:* formal analysis under infrastructure-denied conditions.

F. G6: Scheduler-Aware Threat Models

Only five papers study MAC-layer security, and none model schedulers as security-relevant. Our case study shows scheduler choice changes availability collapse severity under identical stressors. *Target:* wireless security, real-time systems. *Needed:* scheduler-aware threat models.

G. G7: Policy-Safe Configuration Envelopes

A single DL/UL policy change caused $\approx 30\%$ deadline misses, yet no framework defines safe operating envelopes. O-RAN near-RT RIC increases this risk [17]. *Target:* O-RAN, safety engineering. *Needed:* runtime-safe configuration enforcement.

H. G8: Deadline-Miss Benchmarks

No corpus paper reports deadline-miss rate (Table II), preventing reproducible evaluation against operational KPIs. *Target:* wireless security, SA5/SA6. *Needed:* standardised benchmark suites.

I. G9: Cross-Layer Security–Scheduling Co-Design

67% of taxonomy cells contain ≤ 3 papers, and cross-layer availability is absent (Figure 1). PHY mechanisms such as PLA [3] and schedule randomisation [4] interact with MAC scheduling in unstudied ways. *Target:* CPS security, protocol design. *Needed:* end-to-end latency-distribution evaluation.

J. G10: 6G Threat Vectors

RIS, NTN, JCAS, and semantic communications introduce new integrity and availability threats [49], none addressed in TS 33.180 or corpus papers. *Target:* 6G security, SA3. *Needed:* mc-MTC-specific threat analysis.

K. G11: AI/ML Security for mc-MTC

ML-based defences [5], [6] rarely evaluate adversarial robustness or inference latency, despite timing constraints making slow detection unusable. *Target:* adversarial ML, edge computing. *Needed:* joint evasion and availability evaluation.

L. Summary

All gaps share the same structural weakness: defences are evaluated in isolation, under benign timing assumptions, and mainly against external adversaries. Progress requires reporting latency distributions and deadline-miss rates, modelling insider and policy-level threats, and tighter collaboration across formal methods, wireless security, and standards communities.

VIII. CONCLUSION

This paper presents the first systematisation of mc-MTC security across 91 papers and 14 3GPP specifications. Across all analyses, we show that mc-MTC security cannot be evaluated without treating availability, measured through latency distributions and deadline-miss rates, as a first-class property.

Our corpus analysis reveals a major imbalance: 74% of studies focus on standards-defined properties, 73% on external adversaries, yet none report safety-critical deadline-miss rates. Analysis of TS 33.180 further exposes structural weaknesses: 11 of 13 MCX protocol flows lack formal verification, MIKEY-SAKKE requires key escrow, and its post-quantum migration path depends on an RFC dormant since 2012.

We validated these risks experimentally through an ns-3/5G-LENA case study, where standards-compliant stressors increase mean latency by $\approx 3\times$ and trigger $\approx 30\%$ deadline misses at 0.5 ms without cryptographic failure. The system therefore *fails securely*.

These findings highlight the need for availability-centric evaluation, policy-aware threat models, and urgent verification of MCX protocols. As with TETRA, insufficient scrutiny risks embedding systemic weaknesses into future mission-critical deployments.

APPENDIX

Tables VII–X list the papers in the corpus with their five-dimensional classification. Papers spanning multiple security objectives (D2) are listed under each applicable objective. **D1**: PHY, MAC, Core, App, Cross. **D2**: C=Confidentiality, I=Integrity, A=Availability, Au=Authenticity, P=Privacy. **D3**: Ext=External, Ins=Insider, Priv=Privileged, XS=Cross-slice. **D4**: RR=Radio resource, SP=Scheduling predictability, TL=Traffic load, CO=Computational overhead, VB=Virtualisation boundaries. **D5**: SR=Secrecy rate, BL=BLER/PER, DA=Detection accuracy, CF=Formal proof, LD=Latency distribution.

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TABLE VII: Corpus classification (1/4): PHY layer.

#	Paper (Year, Venue)	D1	D2	D3	D4	D5
Finite Blocklength Secrecy						
P1	PLS in Finite Blocklength (2016, IEEE Conf.)	PHY	C	Ext	RR	SR
P2	PLS for URLLC (2019, arXiv/Journal)	PHY	C	Ext	RR	SR
P3	PLS Finite Blocklength over Fading (2020, TIFS)	PHY	C	Ext	RR	SR
P4	PLS Finite Blocklength Slow Fading (2020, IEEE)	PHY	C	Ext	RR	SR
P5	Secure URLLC Resource Alloc. MC IoT (2020, TC)	PHY	C	Ext	RR	SR
P6	Secure Multi-User MISO-URLLC (2020, ICC)	PHY	C	Ext	RR	SR
P7	Tx Power Min. Secure Short-Pkt (2021, IEEE)	PHY	C	Ext	RR	SR
P8	Secure Tx Rate Short Pkts Queueing (2021, TWC)	PHY	C	Ext	RR,CO	SR
P9	Security & Reliability Random Eaves. (2021, IEEE)	PHY	C	Ext	RR	BL
P10	Quantum Learning Coding Secure URLLC (2021, arXiv)	PHY	C,Au	Ext	RR	BL
P11	Optimizing Training & Tx Secure URLLC (2022, IEEE)	PHY	C	Ext	RR	SR
P12	Secrecy Throughput ANECE (2022, IEEE Conf.)	PHY	C	Ext	RR	SR
P13	Impact Mobility PLS 5G URLLC (2022, Wiley)	PHY	C	Ext	RR	SR
P14	Secure Short-Pkt UL Massive MIMO (2023, IEEE)	PHY	C	Ext	RR	SR
P15	Secure Short Pkts Wireless Powered (2023, IEEE)	PHY	C	Ext	RR	SR,BL
P16	Secure URLLC UAV-Enabled NOMA (2024, IEEE)	PHY	C	Ext	RR	SR
P17	Survey PLS Ultra/Hyper RLLC (2024, COMST)	PHY	C	Ext	RR	SR
Jamming & Anti-Jamming						
P18	5G NR Jamming, Spoofing, Sniffing (2018, ICC)	PHY	A,C	Ext	RR	BL
P19	MC Portable LTE in RF Interference (2017, MIL-COM)	PHY	A	Ext	RR	BL
P20	Smart Jamming 5G NR Review (2020, IEEE)	PHY	A	Ext	RR	DA
P21	RT ML Hoeffding Trees Jamming Det. (2021, IEEE)	PHY	A	Ext	RR	DA
P22	AI Channel Awareness 5G Scheduling (2021, IEEE)	PHY	A	Ext	RR,SP	LD
P23	Jamming Det. 5G Bayesian Net (2022, IEEE)	PHY	A	Ext	RR	DA
P24	Adaptive Reserv. Jamming NR-V2X (2022, IEEE)	PHY	A	Ext	RR	BL
P25	Space-Time Jamming Det. MC mMTC (2023, IEEE)	PHY	A	Ext	RR	DA
P26	UE-Assisted Jamming Det. 5G NR (2023, IEEE)	PHY	A	Ext	RR	DA
P27	Intelligent Jamming PBCH 5G NR (2023, IEEE CL)	PHY	A	Ext	RR	DA
P28	Bayesian ML Jamming Det. 5G NR (2023, IEEE)	PHY	A	Ext	RR	DA
P29	5G Jammers Spectrograms CNN (2024, IEEE)	PHY	A	Ext	RR	DA
P30	Smart PRACH Jamming 5G Campus (2024, arXiv)	PHY	A	Ext	RR	BL
Physical-Layer Authentication						
P31	PL Channel Auth 5G via ML (2018, Wiley WCMC)	PHY	Au	Ext	RR	DA
P32	PLA for 5G: Opportunities (2020, IEEE JSAC)	PHY	Au	Ext	RR	DA
P33	Delay Perf. PLA Under Sybil (2021, IEEE)	PHY	Au	Ext	RR	LD
P34	SL for PL Message Auth URLLC (2021, IEEE)	PHY	Au	Ext	RR	DA
P35	Multi-Feature PLA URLLC (2023, IEEE)	PHY	Au	Ext	RR	DA
P36	URLLC PLA Non-linear SL (2023, IEEE)	PHY	Au	Ext	RR	DA

TABLE VIII: Corpus classification (2/4): MAC and Core/Edge layers.

#	Paper (Year, Venue)	D1	D2	D3	D4	D5
MAC Layer						
M1	Graph Coding Pilot Auth GF-URLLC (2020, IEEE)	MAC	Au,A	Ext	SP	BL
M2	Pilot Contamination Det. mmWave GF (2020, TIFS)	MAC	Au,A	Ext	SP	DA
M3	Online Schedule Randomisation (2023, TOSN)	ACM	MAC	A	Ext	SP
M4	Dynamic Proactive-Reactive Sched. (2024, rect)	SDi-	MAC	A	Ext	SP
M5	Capacity 5G NR Grant-Free (2019, MDPI Sensors)	MDPI	MAC	A	Ext	SP
5G-AKA Formal Verification						
C1	Formal Analysis 5G Auth — Basin (2018, CCS)	Core	Au,C	Ext	CO	CF
C2	Component-Based 5G-AKA — Cremers (2019, NDSS)	Core	Au	Ext	CO	CF
C3	5GReasoner — Hussain (2019, CCS)	Core	Au,P,A	Ext	CO	CF
C4	5G-AKA Privacy — Koutsos (2019, EuroS&P)	Core	P,Au	Ext	CO	CF
C5	Generic AKA Construction 5G (2019, IEEE)	Core	Au	Ext	CO	CF
C6	Formal Verif. Primary Auth (2020, IEEE)	Core	Au	Ext	CO	CF
C7	Privacy-Preserving AKA 5G (2021, USENIX Sec.)	Core	P,Au	Ext	CO	CF
C8	Formal Verif. 5G EAP-AKA (2021, IEEE)	Core	Au	Ext	CO	CF
C9	Security Verif. AKA Petri-net (2021, IEEE)	Core	Au,I	Ext	CO	CF
C10	Formal Spec. 5G-AKA mCRL2 (2021, IEEE)	Core	Au	Ext	CO	CF
C11	Formal Analysis AKA Strand Space (2022, MDPI)	Core	Au	Ext	CO	CF
C12	Attacks Security Context 5G (2023, arXiv)	Core	Au,I	Ins	CO	CF
C13	Dependency-Graph 5G AKA (2024, IEEE)	Core	Au,C	Ext	CO	CF
C14	5G Specs Formal Verif. LLMs (2024, IEEE)	Core	Au	Ext	CO	CF
C15	5G-AKA-FS Forward Secrecy (2024, MDPI)	Core	Au,C	Ext	CO	CF
Security Testing & Deployment						
C16	Security & Protocol Exploit 5G (2018, MILCOM)	Core	Au,C,A	Ext	CO	CF
C17	Insecure Connection Bootstrap (2019, WiSec)	Core	Au	Ext	RR	CF
C18	New Vulns 4G/5G Device Caps (2019, WiSec)	Core	P,Au	Ext	RR	CF
C19	Handover Vulns (2021, ACSAC)	Core	Au,C,A	Ext	CO	CF
C20	Formal Analysis Handover (2021, WiSec)	Core	Au,C	Ext	CO	CF
C21	5G SUCI-Catchers (2021, WiSec)	Core	P	Ext	CO	DA
C22	Measuring 5G Security (2022, WiSec)	Core	C,I,Au	Ext	CO	CF
C23	UE Security Reloaded 5G SA (2023, WiSec)	Core	Au,I,C	Ext	CO	CF
C24	European 5G Security Wild (2023, WiSec)	Core	C,I	Ext	CO	CF
C25	Never Let Me Down: Bidding-Down (2023, WiSec)	Core	C,I	Ext	RR	CF
C26	Nori: Concealing ID 5G (2021, ARES)	Core	P	Ext	CO	CF
C27	Fixing Insecure SIBs (2024, RAID)	Core	Au,I	Ext	CO	LD
Network Slicing Security						
C28	Strong 5G Slice Isolation (2018, IEEE)	Core	C,I	XS	VB	CF
C29	Slice Component Association (2018, MSWiM)	Core	Au,P	XS	VB	CF
C30	Secure Slicing: DDoS Mitigation (2019, CNS)	Core	A	Ext	TL	LD
C31	Slicing Security Overview (2020, IEEE Access)	Core	A,C,I	XS	VB	—
C32	Secure5G: DL Framework (2020, IEEE)	Core	A,C	Ext	VB	DA
C33	Mitigate Slice Topology Learning (2021, TDSC)	Core	P,Au	XS	VB	CF
C34	RL-Based Slice Isolation DDoS (2023, IEEE)	Core	A	Ext	TL	DA
C35	5GLatte: Lateral Movement (2023, ITNAC)	Core	C,I	XS	VB	DA

TABLE IX: Corpus classification (3/4): Application and Cross-layer.

#	Paper (Year, Venue)	D1	D2	D3	D4	D5
Application Layer						
A1	Multi-domain Auth MC Services (2016, IEEE)	App	Au	Ext	CO	LD
A2	Route Selection Trust Public Safety (2018, IEEE)	App	I,A	Ins	CO	DA
A3	MIKEY-SAKKE Analysis — Murdoch (2016, IEEE)	App	C,P	Priv	CO	CF
A4	Beyond PS-LTE: Security PPDR (2020, arXiv)	App	Au,C,A	Ext	CO	CF
A5	Security 5G ESSENCE Platform (2021, Springer)	App	C,I,A,Au	Ext	CO	CF
A6	Abusing 5G Warning Systems (2022, ACSAC)	App	I,A	Ext	RR	CF
A7	Privacy Issues MC Communication (2024, arXiv)	App	P	Ext,Ins	CO	CF
A8	MCX RAN Capacity Limitations (2024, IEEE Access)	App	A	Ext	RR	BL
Cross-Layer & Supplementary						
X1	Overview 5G URLLC Security 3GPP (2019, CSCN)	Cross	Au,C	Ext	CO	—
X2	Survey Security 3GPP 5G (2019, IEEE Access)	Cross	All	Ext	CO	—
X3	Post-Quantum Crypto & 5G (2019, WiSec)	Cross	C,Au	Ext	CO	CF
X4	C-V2X Security Requirements (2018, arXiv)	Cross	Au,C	Ext	RR	—
X5	PQ Secure SUCI 6G KEMSUCI (2022, WiSec)	Cross	P	Ext	CO	LD
X6	Beyond-5G AKA Protocol PQ (2022, Springer)	Cross	Au,P	Ext	CO	CF
X7	Enhancing 5G-AKA PQ Sig. (2024, Springer)	Cross	Au	Ext	CO	LD
X8	5G Security URLLC Scenario (2024, IEEE)	Cross	A,Au	Ext	RR	LD
X9	Zero-touch Security mMTC DDoS (2023, IoTJ)	Cross	A	Ext	TL	DA
X10	Securing 5G Control Plane DoS (2021, SDirect)	Cross	A	Ext	TL	LD
X11	Cross-Layer Attack Det. 5G (2024, INFOCOM)	Cross	A,Au	Ext	RR	DA
X12	Unprotected 4G/5G Control (2024, arXiv)	Cross	C,A	Ext	RR	DA
X13	RRC Replay Attack & Securing BS (2022, ARES)	Cross	A,Au	Ext	CO	DA

TABLE X: Corpus classification (4/4): D2D/Sidelink and O-RAN/MEC.

#	Paper (Year, Venue)	D1	D2	D3	D4	D5
D2D / Sidelink Security						
D1	Formal Analysis ProSe AKA Relay (2025, TDSC)	Cross	Au	Ext	CO	CF
D2	Proxy Signature Drone Auth D2D (2021, IEEE)	Cross	Au	Ext	CO	LD
D3	Lightweight Traceable D2D AKA (2021, SDirect)	Cross	Au,P	Ext	CO	CF
D4	D2D Group Auth 5G AKA (2021, SDirect)	Cross	Au	Ext	CO	CF
D5	DDSec/DDACap D2D ProSe (2021, MDPI)	Cross	Au,P	Ext	CO	CF
D6	Secure D2D Massive MTC (2021, SDirect)	Cross	Au	Ext	CO	CF
O-RAN & MEC Security						
R1	Open RAN Security Challenges (2022, arXiv)	Cross	All	Ext,Ins	VB	—
R2	E2E O-RAN Security Arch. (2023, arXiv)	Cross	C,I,A	Ext	VB	CF
R3	Implementing Security O-RAN (2023, arXiv)	Cross	C,I	Ext	CO	LD
R4	MEC-enabled 5G Security Survey (2021, CSUR)	Cross	All	Ext	VB	—
R5	AI-Driven IDS O-RAN 5G (2024, MobiCom)	Cross	A	Ext	VB	DA